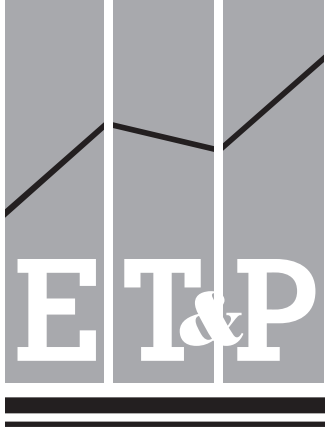




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Explorative and Exploitative Learning from External Corporate Ventures

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This study examines the antecedents of explorative and exploitative learning of technological knowledge from external corporate ventures. We compare different forms of external corporate venturing, namely corporate venture capital investments, alliances, joint ventures, and acquisitions, as alternative avenues for interorganizational learning. Furthermore, we test the effects of multiple relational characteristics on the type of learning outcomes. Our empirical analysis is based on citations in patents filed by a sample of 110 largest U.S. public information and communications technology companies during the years 1992–2000. We find that corporate venturing mode and technological relatedness have significant effects on the likelihood of explorative learning.

Corporate entrepreneurship has emerged as an important subfield in entrepreneurship. Increasingly, it is recognized that entrepreneurial activities are not confined to the domain of small- and medium-sized firms or start-ups but that entrepreneurship is taking place as well in larger, more established corporations (e.g., Ahuja & Lampert, 2001). In fact, several scholars argue that corporate entrepreneurship is central to the survival and renewal of established corporations (e.g., Guth & Ginsberg, 1990; Stopford & Badenfuller, 1994). Corporate entrepreneurship initiatives allow the firm to learn about new technologies, new markets, and new ways to operate (Ahuja & Lampert, 2001; Birkinshaw, 1997; Dess, Ireland, Zahra, Floyd, Janney, & Lane, 2003) all activities critical for the long-term survival and competitiveness of the corporation. Corporate ventures are often vehicles to explore new development avenues for the corporation through radical learning and are also used to exploit existing knowledge through incremental learning. This skill is important in the light of rapid and often dramatic change in firm environments (Eisenhardt & Brown, 1997). High velocity change results in a constant need to adapt and learn new skills but at the same time it also shortens the time given to exploit previously built skills and capa-

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bilities forcing incumbents to excel at both exploration and exploitation (e.g., Eisenhardt & Brown, 1997).

Companies are increasingly using corporate venturing to learn from knowledge sources beyond the boundaries of the firm. Different modes of external corporate venturing (Keil, 2002; Miles & Covin, 2002; Sharma & Chrisman, 1999), such as, corporate venture capital (CVC) investments, alliances, joint venture alliances and acquisitions of entrepreneurial ventures, enable companies to learn. External venturing may enable firms to monitor the development of markets and technologies (Keil, 2002; McNally, 1997; Winters, 1988), to assimilate technologies previously used by their alliance partners (Mowery, Oxley, & Silverman, 1996; Stuart & Podolny, 1996), to enter and expand into new emerging market segments (Barkema & Vermeulen, 1998; Mitchell & Singh, 1992), and more generally to become more innovative and grow faster (Powell, Koput, & Smith-Doerr, 1996; Stuart, 2000).

A large number of studies have empirically analyzed the learning implications focusing on different types of external corporate ventures. For instance, prior studies have shown a positive relationship from CVC investments to parent firm's innovativeness (Dushnitsky & Lenox, 2002) and recognition of technological discontinuities (Maula, Keil, & Zahra, 2003). Alliance research has demonstrated knowledge flows and learning among collaborating companies (Iansiti, 2000; Rosenkopf & Almeida, 2003; Rothaermel, 2001a, 2001b). Research on acquisitions has found a similar relationship between acquisitions and organizational learning (Ahuja & Katila, 2001; Vermeulen & Barkema, 2001). However, these studies have typically focused on the learning implications of a single governance mode thus ignoring the portfolio perspective and comparison of the different types of relationships within the corporate portfolio of external ventures (Salk & Simonin, 2003).

Other studies have shown the effects of various dimensions of an alliance or a joint venture relationship's influence on interorganizational learning. For instance, Lane and Lubatkin (1998) argue that similarity along the dimensions of relatedness of firms' knowledge base, organizational structures, policies, and the dominant logic influence interorganizational learning. Hamel (1991) finds that partners' intent, transparency, and receptivity are important influences on learning in relationships. Yli-Renko, Autio, and Sapienza (2001) find that social interaction with customers relates positively to knowledge acquisition by intensifying interactions, by enhancing the firm's ability to recognize and evaluate pertinent knowledge, and by increasing the incentive to exchange and process information.

The above studies have added very much to our understanding of learning from external sources. However, they share one shortcoming. Their focus is largely on the questions if firms learn and if so how much. In this article, we extend these studies by focusing on the type of learning outcomes through external corporate ventures. In particular, we study how different relationship characteristics relate to explorative or exploitative learning of technological knowledge from corporate venturing partners. We develop an original empirical setting based on patent citation data to test our hypotheses.

The concept of explorative and exploitative learning (March, 1991) contrasts entrepreneurial search for new technological and business opportunities and ways to capture those opportunities with adaptive and more risk-averse learning that leverages existing knowledge. Exploration activities include search, variation, risk taking, experimentation, play, flexibility, discovery, or innovation (March, 1991). Knowledge generated by such activities is often distant from the existing knowledge base of the firm (Katila, 2002). Exploitative learning is a directed search emphasizing limiting variety (McGrath, 2001). Exploitation activities include refinement, choice, production, efficiency, selection, implementation, and execution (March, 1991). Organizations search by using knowledge

that is closely related to their preexisting knowledge bases (Helfat, 1994; Stuart & Podolny, 1996). Despite the seminal nature of March's (1991) distinction and the frequency with which it is cited in the literature, surprisingly few studies have attempted to distinguish between explorative and exploitative learning outcomes in large-scale empirical studies of interorganizational learning.

In addition to empirically examining a new dimension of explorative and exploitative learning, our article focuses on two relational dimensions of collaboration. We develop and test hypotheses concerning the effects of different external venture governance modes and the industry relatedness, the downstream vertical relatedness, and the technological relatedness of ventures on the type of the learning outcomes. Venture relatedness has been studied as an important factor influencing financial performance of external corporate ventures (e.g., Thornhill & Amit, 2001) but has only recently begun to be explored as an influence on corporate learning in external ventures (Autio, Parhankangas, & Sapienza, 2004).

Our study makes three main contributions. First, we contribute to corporate entrepreneurship literature by developing an interorganizational learning perspective to the study of external corporate ventures. Second, we contribute to interorganizational learning literature by empirically examining factors influencing explorative versus exploitative interorganizational learning outcomes in corporate venturing activities. Third, we extend the prior literature by empirically analyzing the effects of different forms of collaboration together with the industry relatedness, downstream vertical relatedness, and the technological relatedness of external ventures. We hope to contribute to a better understanding of factors influencing interorganizational learning.

This article proceeds as follows. First, we briefly review literature pertaining to explorative and exploitative corporate learning and different modes of external corporate venturing. Next, we develop hypotheses regarding the different relationship dimensions and explorative or exploitative learning. We then introduce our sample and methodology, followed by the results of our analysis. Finally, the article concludes with a discussion of the findings.

Explorative and Exploitative Corporate Learning

In his seminal article, March (1991) argues that firms need to balance between activities that contribute to exploration of new knowledge, technology, and capabilities and activities that contribute to exploitation of the existing knowledge or capability base of the firm. Change in the competitive environment creates a constant need to adapt and learn new skills. Yet at the same time, it also shortens the time given to exploit previously built skills and capabilities forcing organizations to strike a balance between maintaining stability and efficiency while adapting to rapid and often unpredictable change (e.g., Eisenhardt & Brown, 1997).

Striking this balance is inherently difficult since exploration and exploitation activities exhibit stark differences. Explorative learning emphasizes learning by generating variation (McGrath, 2001). Therefore, knowledge generated by exploration activities is often distant from the existing knowledge base of the firm (Katila, 2002), for instance, involving technologies the firm has little previous experience in. In contrast, exploitative learning is a directed search emphasizing limiting variety (McGrath, 2001) and building closely on the existing knowledge base. By decreasing variance the short-term mean performance is improved (McGrath, 2001), but often long-term performance is decreased. In March's words: "Compared to returns from exploitation, returns from exploration are

systematically less certain, more remote in time, and organizationally more distant from the locus of action and adaptation. . . . The certainty, speed, proximity, and clarity of feedback ties exploitation to its consequences more quickly and more precisely than is the case with exploration” (March, 1991, p. 73).

Due to these differences in exploration and exploitation activities, many firms tend to optimize exploitation (Benner & Tushman, 2002). Organizational factors cause firms to prefer “true and tried” solutions over novel ones even in highly unusual and unforeseen circumstances (Cyert & March, 1963). Power-political incentives to preserve the established organizational power balance and the calcification effect of emergent organizational routines tend to favor previously used solutions. In addition to those mentioned above, the advances in learning and knowledge-based research on the firm have highlighted important cognitive and learning-based mechanisms that tend to bias firms toward local optimization and local learning. Firms learn efficiently from local and proximate sources because previous related knowledge facilitates the identification, interpretation, and assimilation of related knowledge (Cohen & Levinthal, 1989, 1990, 1994; McGrath, 2001), because new knowledge is often created through the combination of existing knowledge items (Kogut & Zander, 1992), and because related new knowledge is less likely to conflict with the established dominant logic and the established mental models within the firm, thereby facing less resistance against adoption (Levinthal, 1997).

To balance exploration and exploitation, some authors have suggested maintaining a set of organizational activities, each of which contributes to a particular type of corporate renewal in the exploration–exploitation continuum (Burgelman & Sayles, 1986; Keil, 2002). Prior research has suggested that the larger the difference between the focal company and the venturing partner, the lower the integration between the partners should be (Roberts & Berry, 1985). However, while there is plenty of managerial advice on how firms should manage exploration activities (e.g., Burgelman & Sayles, 1986), and increasing literature on organizational cognition and learning, there is surprisingly little empirical research focusing explicitly on the learning impacts of various organizational arrangements for innovation in established firms. In addition, much of the received literature on explorative learning focused on exploration activities that are carried out in-house (Eisenhardt & Martin, 2000; McGrath, 2001; March, 1991; Zahra, Ireland, & Hitt, 2000), thereby sidestepping an increasingly important exploration activity for established firms, that of external corporate venturing.

External Corporate Venturing

The corporate entrepreneurship label has been attached to a wide variety of phenomena ranging from “established” organizations entering a new business, individuals or groups championing innovations within a corporate context, entrepreneurial efforts that renew the organization by changing its business, structure, or strategy, and situations in which an “entrepreneurial philosophy permeates an organization.” Building on the recent work to clarify these concepts (Keil, 2002; Miles & Covin, 2002; Sharma & Chrisman, 1999), we refer to external corporate venturing as the creation of new businesses by corporations in which a corporation leverages external partners in an equity or nonequity interorganizational relationship. Corporations enter into interorganizational relationships either to create new ventures or to develop existing internal business. External corporate venturing activities are a subset of the broader set of entrepreneurial activities that have been referred to in corporate entrepreneurship research (e.g., Covin & Miles, 1999; Sharma & Chrisman, 1999). It complements other corporate entrepreneur-

ial efforts such as internal corporate venturing, i.e., the creation of new businesses within the corporate organization or initiatives geared towards innovation and strategic renewal. These efforts are not inherently alternatives but rather complement each other.

Corporations utilize several governance modes to conduct external corporate venturing. Governance modes include CVC investments, nonequity alliances for the development of new business ventures, joint ventures, and acquisitions of entrepreneurial ventures. In recent years, a large number of companies have utilized some form of venture capital investments (Birkinshaw, van Basten Batenburg, & Murray, 2002; Maula & Murray, 2002). In this form of corporate venturing, the corporation participates in the private equity market imitating the setup and behavior of traditional venture capital firms. Some corporations carry out CVC investments through their own, usually autonomous, investment unit or by participating in a fund managed by a traditional venture capital firm. Goals of CVC include financial gain and potential strategic benefits such as learning of markets and technologies, or market creation (Chesbrough, 2002). Interaction between the venture and the corporation is usually mediated by the CVC unit or through the venture capitalist managing the fund the corporation participates in. This intermediation differentiates CVC investments from direct minority investments. Direct minority investments are usually carried out by a business unit of the corporation while CVC investments normally utilize a dedicated investment organization or invest through a traditional venture capital firm (e.g., Miles & Covin, 2002). However, corporate direct minority investment activities and CVC have much in common in that they both rely on investment in the partner as a basis for the relationship.

Venturing through nonequity alliances aims at the development of new businesses together with an external partner. In contrast to direct minority investments and joint ventures, nonequity alliances are based on contracts and do not use ownership in the relationship. This form of relationship is used to develop existing internal ventures or create a framework for joined business development. It needs to be distinguished from alliances for other objectives such as supplier–customer alliances or distribution alliances since the management of alliances exhibits dramatically different characteristics (Doz & Hamel, 1998).

Joint ventures refer to partnerships in which a new legal entity and organization is formed by two companies to pursue a business opportunity together. While minority investments are widely used with small and medium companies, joint ventures are more relevant for relationships with larger, more established firms. Joint ventures differ from direct minority investments because companies form a separate organization.

In an acquisition of an entrepreneurial venture, an independent venture or an external venture is internalized by the acquisition of the majority of the share capital of a venture. An acquired venture may then be operationally integrated into the acquiring company, but it may also continue as a relatively independent company (Laamanen, 1997). Similar to alliances, acquisitions of entrepreneurial ventures should be distinguished from other acquisitions such as mergers or acquisitions to consolidate an industry that exhibit different characteristics (Bower, 2001).

Hypothesis Development

Governance Modes and Explorative and Exploitative Learning

Different governance modes to conduct external corporate ventures are likely to differ in the degree to which they support explorative and exploitative learning. We argue that these differences are produced not only by differences in how well different modes

support transfer of knowledge but also by the varying cost and time associated with managing the ventures.

First, governance modes differ in the degree of integration between the partners (e.g., Burgelman, 1984; Roberts & Berry, 1985). Corporate venture capital investments are the most arms-length relationships among the four governance modes we study. In CVC investments, the corporation interacts with the venture it invests in mainly through a dedicated, often separate, CVC operation (Keil, 2002). Nonequity venturing alliances are typically formed between a business unit and a new venture. Since cooperation takes place directly with a business unit, integration is stronger than in CVC investments. Joint ventures include the formation of a dedicated organization jointly owned and in many instances also jointly staffed by the venture partners therefore providing for stronger integration than the previous two governance modes. Acquisitions of entrepreneurial start-ups, the fourth governance mode, lead to the closest integration. Close integration is an important prerequisite for social interaction, which many studies have found to facilitate learning (Bresman, Birkinshaw, & Nobel, 1999; Kale, Singh, & Perlmutter, 2000; Kogut & Zander, 1992; Mowery et al., 1996; Steensma, 1996; Yli-Renko et al., 2001).

The above argument might suggest that more tightly integrated governance modes, such as joint ventures or acquisitions, could better support explorative learning. When conducting explorative learning, firms lack the underlying often tacit knowledge to build on. However, we argue that several effects outweigh the effects of this increased interaction when considering the type of learning outcomes. First, although integration might create an opportunity for improved interaction, integration also reduces flexibility to adapt to change. Explorative learning, as defined by March (1991), requires more flexibility and adaptation, and lacks the certainty, speed, proximity, and clarity of feedback that are characteristic for exploitative learning. Therefore, less integrated external venture governance modes may provide a more effective environment for discovery of new knowledge derived from their greater flexibility and adaptability to change (Osborn & Hagedoorn, 1997). Joint ventures, and to a larger extent acquisitions, are more difficult to adapt to changes in the environment.

The uncertain nature of the returns to explorative learning (March, 1991) and the unknown strategic importance and operational relatedness of ventures geared toward explorative learning (Burgelman, 1984) might lead to a situation where corporations would choose less integrated governance mechanisms, such as CVC investments or nonequity venturing alliances, for projects that are explorative in nature. This argument is in line with transaction cost reasoning (Williamson, 1975, 1985). For a given project that carries a given uncertainty, a firm should reduce investments in assets that cannot be redeployed. An acquisition of an entrepreneurial venture entails the most significant investments into relationship-specific assets, followed by joint ventures, nonequity venturing alliances, and CVC investments. Corporate venture capital investments entail less investment into relationship-specific assets than nonequity alliances, since these relationships are entered into for a financial objective beyond the strategic learning objective. Taken together we hypothesize:

Hypothesis 1a: Corporate venture capital investments are more likely to lead to explorative (versus exploitative) learning than nonequity venturing alliances.

Hypothesis 1b: Nonequity venturing alliances are more likely to lead to explorative (versus exploitative) learning than joint ventures.

Hypothesis 1c: Joint ventures are more likely to lead to explorative (versus exploitative) learning than acquisitions.

Venture Relatedness and Explorative and Exploitative Learning

Venture relatedness has been an important variable in both interorganizational learning and diversification literature. External corporate ventures can be related to the focal corporation along several dimensions. In this study, we focus on industry relatedness, downstream vertical relatedness, and technological relatedness.

Industry relatedness. Ventures can be related to the corporation by operating in related industries. The more related an industry is, the more we expect learning to be incremental, building, and refining existing knowledge. The basic argument is that related industries share important characteristics that support incremental learning and make explorative learning less likely. Industries can be related along a large number of dimensions. Previous studies have operationalized industry relatedness using products and services the industries offer, inputs they use, the customers they serve, the technologies they use, the way they organize operations, the cognitive frameworks that drive executive decisions (see e.g., Farjoun, 1994; Prahalad & Bettis, 1986; Robins & Wiersema, 1995; Silverman, 1999). The more closely related two industries are, the more similar firms should be along these dimensions. This similarity should allow two firms to exchange knowledge more easily. Similarity in business logic and cognitive frameworks make it also more likely that firms will share the same views of the future evolution of markets and technologies and will search for new knowledge in similar places. External corporate venture involving firms from closely related industries are therefore more likely to be focused on knowledge close to the existing knowledge base of the firm. The similarity of knowledge and strategy agenda should make it more likely that knowledge exchanged is an extension and refinement of what the firm already knows thus biasing learning toward exploitative learning. We hypothesize

Hypothesis 2: Industry relatedness is negatively related to explorative (versus exploitative) learning.

Downstream vertical relatedness. In their study of the world disk drive industry, Christensen and Bower (1996) show that the demands of a firm's customers shape the allocation of resources in technological innovation. Because of the pressure to serve the current business interests of the customers, established firms may focus their development on technologies that build on existing technologies their customers have deployed at the expense of exploring more distant technologies that might not be immediately applicable with their current customer base. External corporate ventures that are carried out jointly with customers are likely to be driven by the interests of the customers to receive improvements to the products and platforms they already have invested in, rather than induce the supplier to explore radical new solutions that might make the prior relationship-specific investments obsolete.

Aside from the pressure of customer interest, a previous customer relationship also suggests convergence of knowledge bases. As a result of working together, the relevant technological bases of the customers and suppliers are likely to be aligned thus suggesting incremental, exploitative learning rather than radical, explorative learning.

Hypothesis 3: Downstream vertical relatedness is negatively related to explorative (versus exploitative) learning.

Technological relatedness. In Hypothesis 2, we predicted that overall industry relatedness of the partners in an external corporate venture makes exploitative learning more

likely. While industry relatedness should lead to an overlap in the knowledge base of the firm, firms within an industry might also exhibit dramatic differences in their technology base. For instance, firms might utilize competing technologies or different business models. Therefore, we argue that a stronger predictor of the likelihood of exploitative learning is the technological relatedness of two firms.

Technological relatedness has been extensively studied as a predictor of the amount of learning in interorganizational relationships such as alliances (Lane & Lubatkin, 1998), acquisitions (Ahuja & Katila, 2001) and joint ventures (Shenkar & Li, 1999). Building on Cohen and Levinthal's (1989) argument, the above studies argue that a firm's previous, related knowledge facilitates the identification, interpretation, and assimilation of related knowledge. Learning should therefore increase if the knowledge bases of the two partners are similar (Lane & Lubatkin, 1998). More recently, some authors have refined this argument by pointing out that a too large similarity in knowledge bases reduces learning since little new is to be learned (Ahuja & Katila, 2001; Autio et al., 2004; Shenkar & Li, 1999).

While these studies focus on the amount of learning, their argument can form the basis for predictions of the relationship between relatedness and explorative and exploitative learning. With very closely related knowledge bases, mainly exploitative learning takes place. Since both partners share most of each others knowledge, knowledge transferred is likely to be refining or incrementally extending the existing knowledge base of the firm. While learning from unrelated knowledge sources is more difficult, what is learned is more likely new to the firm. The less related the knowledge base of the partner in an external corporate venture, the more likely the learning of the corporation will be explorative. Therefore, we hypothesize:

Hypothesis 4: Technological relatedness is negatively related to explorative (versus exploitative) learning.

Methods

Sample and Data Collection

Our sample was drawn from external corporate ventures of the 110 largest public U.S. corporations operating in four sectors of information and communications technology (ICT) industries: computer equipment, electronics, software, and telecommunications. In this research, we will subsequently refer to these 110 companies as *focal companies* to distinguish them from their external venturing partners, referred to as *partner companies* in this research. We selected the largest companies operating in the ICT sectors based on their revenue in year 1989. We collected the focal companies' financial measures for years 1989–2001 from *Compustat* database. Data of the external ventures was drawn from Thompson Financial's Platinum database. This data included all CVC investments, alliances, joint ventures, and acquisitions made by the focal companies during years 1990–2000. After checking the data for duplicates and incomplete records, we found a total of 8,176 external ventures with over 4,000 different companies. We collected U.S. patents for the focal companies and all their partners using Derwent's World Patent Index, covering patents filed during years 1989–2000. The focal companies filed 5,091 patents that cited a patent of their external venture partner, (61% of which were considered explorative according to our criteria). These patent citations form the basis of our empirical analyses.

Dependent Variable: Explorative (versus Exploitative) Interorganizational Learning

In our empirical analyses, we contrast exploitative and explorative learning outcomes from external ventures. More specifically, we track how, in their inventions, focal companies utilize and further develop technological solutions invented by their collaborators. Our choice to focus on interorganizational learning of technology is motivated by four factors. First, we assumed that the tension of exploitative and explorative learning would be strongly present in product development and in technological learning more generally (Knott, 2003). Second, technological learning was likely to be crucial for every company in the fast-moving ICT industries. Third, prior studies of interorganizational learning have often focused specifically on technology-related learning, so our choice enables us to compare and contrast our findings to previous research (e.g., Almeida, Dokko, & Rosenkopf, 2003; Lane & Lubatkin, 1998; Mowery et al., 1996). Fourth, we are able to operationalize technological learning through patent data and avoid problems related to survey research.

To measure events of interorganizational learning, we analyze patent citations. Patents establish property rights to knowledge gained from entrepreneurial (or other) activities and thereby protect it for later appropriation. As part of the patent, a firm has to provide citations to other patents. These citations serve the function of delineating the new knowledge in a patent from what has been known previously thereby limiting the scope of the patent. A citation of a patent A by a patent B means that A contains knowledge upon which B builds. A citation from patent A to patent B means that the patent examiner verifies citations and may include additional citations to relevant prior patents if deemed necessary (Podolny & Stuart, 1995).

Patent citations have been widely used in prior literature to measure knowledge flows between companies and geographical areas (e.g., Ahuja & Katila, 2001; Jaffe, Trajtenberg, & Henderson, 1993; Mowery et al., 1996; Rosenkopf & Nerkar, 2001; Stuart & Podolny, 1996). For instance, Jaffe, Trajtenberg, and Fogarty (2000) surveyed inventors and found that direct patent citations are a relatively good indicator of knowledge flows. The strength of patents is that they are systematically compiled, have detailed information, and are continuously available across time (Rosenkopf & Almeida, 2003). The most common sources of critique are the large variance in the value of patents and the propensity to patent. The likelihood that an innovation is patented depends on the nation, industry, and even company-specific practices and policy. Patents might represent therefore only a portion of innovations made by a company, and the number of patents a company has filed does not necessarily indicate the value or the quantity of their innovative output (Desrochers, 1998). Patent-citation-based measures might also suffer from noise since some citations might not reflect organizational learning but might be made to protect a firm from litigation (Rosenkopf & Almeida, 2003). Despite these weaknesses, patent citations as a measure for organizational learning are among the most frequently used measures for interorganizational learning and allow us to observe the overall technological knowledge flows in our sample.

In our context, every patent filed by our focal companies that makes reference to patents held by their venturing partners is considered an interorganizational learning event in which the focal firm learns knowledge from the venturing partner. Following the same logic, patents that do not make reference to patents of a venture partner are excluded from the analysis since they do not constitute *interorganizational* learning. We therefore first selected all patents filed by the focal companies during years 1992–2000 that cited an invention made by a venture partner (the lower half of the matrix in Figure 1).

Figure 1

Operationalization of the Dependent Variable: Explorative Versus Exploitative Learning from External Ventures

		Citations to own prior patents	
		No	Yes
Citations to partner's patents	No	Source of knowledge not known	
	Yes	Explorative interorganizational learning	Exploitative interorganizational learning

Among interorganizational learning events, we distinguish between explorative and exploitative learning events. Citations to own patents indicate that prior own knowledge is being expanded, refined, or leveraged. In other words, citations to own patents reflect exploitative learning. Thus, we classify all patents with citations to prior patents of the focal company as exploitative. We only classify such patents as explorative that do not cite prior patents of the focal firm and therefore do not build on prior knowledge of the focal firm. Our dependent variable, *explorative learning*, is a binary indicator measuring the absence or presence of citations to own prior patents. The variable takes value 1 if there is no citation to a prior own patent. The variable takes the value 0 if there is a citation to an own prior patent.

Although patents may cite a number of own patents, we believe that a binary indicator of citations of own prior patents has higher validity compared to potential alternative specifications such as the percentage of patent citations referring to own patents in a patent citing a patent of the partner. It is widely recognized that the value and scope of patents vary significantly (Desrochers, 1998), and technological developments may be protected by a varying number of related patents. Our binary operationalization is more robust against such disturbances. However, to ensure the robustness of our results, we constructed an alternative measure indicating the percentage of patent citations referring to own patents and calculated the models using ordinary least squares regression model. This model yielded similar but even stronger results compared to those reported in this article.

We made several limitations regarding the inventions included in the sample. Alliances and joint ventures seldom last more than 5 years, and announcements related to termination of alliances are rarely given (Ahuja, 2000). Likewise, the effects from acquisitions are likely to fade within 5 years, since learning between the acquirer and target becomes an internal knowledge flow. Therefore, we only included patents where

companies had announced collaboration or acquisition within 5 years preceding the patent filing.¹ In case of very large partners, it may be problematic to link learning outcomes to alliances or joint ventures between the companies. For example, there appeared to be thousands of cross-citing patents between Sun Microsystems and Microsoft. These large companies would be likely to cite each other's patents simply because they have such a large number of inventions in the same area. To limit the noise from knowledge transfer outside external ventures, we chose to exclude collaborative agreements among the very largest companies. We thus removed observations if the venturing partner had filed more than 100 patents during 3 years prior to the patent filing.² This limitation is also in line with our focus of external venturing. Given that we do not have other measures for the size of the thousands of external ventures, we used the number of patents as a rough proxy of the size and age of the venture because of the typically high correlations between the number of patents and the size and the age of the firm (e.g., Hall & Ziedonis, 2001; Sorensen & Stuart, 2000).

Our operationalization avoids some of the above-mentioned critiques of patent-citation-based measure by focusing on the nature of patents instead of their quantity and by avoiding problematic cross-national and cross-industry comparisons. In addition, we control for prior firm-level tendency for self-citing, mitigating bias resulting from heterogeneous organizational practices that affect the patenting process. Broad use of citations in U.S. patents support our way of measuring relatedness to prior innovations. Unlike European and Japanese patent offices, U.S. Patent and Trademark Office generally includes citations to a broad range of relevant prior patents beyond the most crucial inventions (Michel & Bettels, 2001). Furthermore, ICT is a good industrial setting for patent-based studies, since companies operating in these industries actively apply for patents (Cohen, Nelson, & Walsh, 2000). While there may be differences in citation practices that depend on the examiner reviewing the patent application (Cockburn, Kortum, & Stern, 2002), we believe that differences based on patent examiner are likely to be randomly distributed. To the extent propensity for self-citing is tied to a specific geographic area, it is mitigated by including the share of explorative previous patents in the regression models.

Independent Variables

We created four binary variables to distinguish between different external venturing modes. *Corporate venture capital*, *alliance*, *joint venture*, and *acquisition* indicate whether the latest form of collaboration between companies is of the particular type. Only the first three dummies are entered into the regression models to avoid perfect collinearity. In regression models, the effects of acquisitions are included in the constant term.

We constructed the variable *industry relatedness* based on the primary Standard Industry Codes (SIC) of the companies. Industry relatedness variable was based on the four-number primary SIC codes of the focal company and its partner. The index received values 0, .25, .333, .5, or 1 depending on the number of matching first digits in the primary SIC codes. The same or very similar measures have been commonly used in prior research

1. To test the robustness of this choice, we also tested alternative models using patents that had been filed within 2, 3, and 4 years after the latest collaborative agreement. The results were effectively the same as those reported here.

2. We also tested alternative limitations 50, 500, and 1,000 without substantial changes in the results. This limitation dropped about 70% of observations from the analysis.

(Finkelstein & Halebian, 2002; Halebian & Finkelstein, 1999; Villalonga & McGahan, 2003).

Since we could not directly determine whether companies had customer relationship with one other, we measure the *downstream vertical relatedness* as the share of sales from the primary industry of the focal firm to the primary industry of the partner company, based on U.S. industry input–output tables (Fan & Lang, 2000). This measure represents the extent to which the partner is a typical buyer of products manufactured or developed by the focal company.

The *technological relatedness* of collaborating companies is based on patent classes the companies filed patents in. We counted the number of patents in each patent subclass using International Patent Classification coding. The patent count was weighted by the number of subclasses in each patent. We used a moving 3-year window to construct patent class counts to reflect the focus of recent technological activities. We then calculated the relative overlap of two companies' patent portfolios using the following formula (adopted from Podolny, Stuart, & Hannan, 1996; Stuart & Podolny, 1996).

$$Tech\ Relatedness = \frac{\sum_k \sqrt{C_{k,A} * C_{k,B}}}{\sqrt{Patents_A * Patents_B}}$$

We multiplied the number of patents in patent classes (k) for companies A and B, summed the results from every patent class, and then divided the result by the geometric mean of patent portfolio sizes. This formula gives each company pair an index between 0 and 1, representing the similarity of the companies' patent portfolios. In this index, a value of 0 corresponds to no overlap between patent classes and 1 corresponds to perfect overlap in patent classes that the focal firm and its partner patented in. The index for each company pair was calculated yearly so that the technological relatedness measure for each observation is based on patents filed during the 3 preceding years.

Control Variables

We utilize a range of control variables to limit the risk of omitted variable bias. Dummy variables for years 1993–2000 are included to control for effects caused by economic cycles, using year 1992 as the default in our regression model. We also include industry dummies, based on the three first digits of the SIC, to account for industry-wide differences in explorative and exploitative learning. Furthermore, we include natural logarithm of sales to account for size of the company, and the percentage of *research and development intensity* to take into account the variance in inputs for innovation activity. We also include *target patents*, the count variable of patents granted to the partner company during the previous 3 years, to account for possible effects caused by the size of the partner company.

To account for heterogeneous company-specific organizational routines, we constructed a variable that indicates the share of explorative inventions in all the previous observation for the company, *previous explorative*. We expect this variable to explain a large share of firm-level heterogeneity in share of explorative inventions, thus strengthening the robustness of our findings.

Patents cite a varying number of prior inventions, depending on the scope of the patent. Patents with a large number of citations may also be more likely to cite a patent made by the focal company. To distinguish between the factors that affect the size of inventions resulting from factors affecting the explorative and exploitative learning, we added a control *patent scope*. The variable indicates the number of citations to prior

patents. While the scope of the patent increases the likelihood that it cites a patent made by the focal company, the diversity of a company's technological portfolio is likely to affect the explorativeness of subsequent learning. Companies with broadly diversified technological portfolios appear more likely to follow an explorative strategy. We thus followed prior research and calculated the Herfindahl index of patenting within different patent classes, *technological diversification* (e.g., Fleming, 2002).

Finally, we control for the number of times the focal company has cited the specific partner's patent, represented by the variable *previous citations to patent*. This takes into account the possibility that partner's knowledge creates a stream of patents, in which the focal company's subsequent patents cite the first patent. According to our specification, these cases would create exploitative learning after the first patent because they start to build on company's prior knowledge—albeit prior knowledge that has only recently been acquired.

Statistical Methodology

Since our dependent variable is dichotomous, we use logistic regression to test our hypotheses. There is varying number of inventions for each corporate venturing arrangement in our data, thus causing possible nonindependence of the observations. We used the statistical program STATA that enables us to cluster observations based on the focal company–partner company dyad. Using this method, observations within each venturing relationship are considered interdependent and our standard deviations are not lowered because of multiple observations within each external venture. This provides us robust estimates of variance (StataCorp, 2000). The results were effectively the same when we clustered observations based on each focal company.

Our methodology creates more observations for those collaborative agreements that have a high level of interorganizational learning. In effect, we explain the type of the learning benefits. We argue that this is desirable and that it also makes more sense to observe the type of learning outcomes in collaborative relationships with a relatively high degree of knowledge transfer. Our robustness tests show that our hypothesized effects are more likely to be present in collaborative agreements with high levels of learning.

Results

Table 1 summarizes the descriptive statistics and correlations for our sample. Each observation is a citation by a focal company's patent to a patent of a venturing partner. The correlation coefficients between independent variables are relatively low, except for alliances and joint ventures (−.68) and downstream vertical relatedness and industry relatedness (.76). Only 35 observations (patent citations) out of 5,091 are associated with CVC investments. Companies that CVC investments are made in tend to be small start-ups that have very few or no patents. This limits the potential to observe learning from CVC investments with patent-citation-based methodology.

Results for logistic regression models are shown in Table 2. We first briefly review the results in Models 1–3 and then comment on our hypotheses based on Model 4, which includes all the independent variables. The effects of the control variables on explorativeness of learning outcomes in external venturing are shown in Model 1. As expected, the repeated utilization of the prior invention is positively related to the likelihood that it will be utilized in exploitative manner. Likewise, observed patent's scope, defined as the overall number of patent citations in the patent, increases the likelihood that the

Table 1

Correlation Coefficients and Descriptive Statistics (N = 5,091)

	Mean	Std	Min	Max	1	2	3	4	5
1 Exploration	.61	.49	.00	1.00					
2 Corporate venture capital	.01	.08	.00	1.00	.00				
3 Alliance	.74	.44	.00	1.00	.14	-.14			
4 Joint venture	.14	.35	.00	1.00	.01	-.03	-.68		
5 Acquisition	.11	.32	.00	1.00	-.20	-.03	-.61	-.14	
6 Industry relatedness	.39	.39	.00	1.00	-.09	-.02	-.16	-.09	.32
7 Downstream vertical relatedness	.04	.04	.00	.13	-.09	-.02	-.20	-.07	.36
8 Technology relatedness	.04	.04	.00	.30	-.17	-.04	-.15	-.10	.33
9 Patent scope	21.72	32.28	1.00	764.00	-.18	.00	-.06	.00	.09
10 Previous citations to patent	5.07	18.30	.00	172.00	-.19	.00	-.24	-.09	.44
11 Previous explorative (%)	.67	.19	.00	1.00	.23	.03	.02	.04	-.08
12 Sales (natural logarithm)	9.26	1.47	4.30	11.31	-.09	.01	.16	-.01	-.20
13 Research and development spending (%)	.10	.05	.00	.32	-.03	.01	-.18	-.01	.25
14 Technological diversification	.26	.19	.06	1.00	.08	-.04	-.07	.04	.06
15 Target patents	29.70	27.96	.00	99.00	.05	-.07	.05	.12	-.18
	6	7	8	9	10	11	12	13	14
7 Industry forward integration	.76								
8 Technology relatedness	.04	.17							
9 Patent scope	.02	.00	-.03						
10 Previous citations to patent	.25	.13	.23	.04					
11 Previous explorative (%)	-.12	-.08	-.10	-.02	-.12				
12 Sales (natural logarithm)	-.17	-.12	-.02	-.12	-.18	-.20			
13 Research and development spending (%)	.14	.15	.14	.01	.32	.01	-.54		
14 Technological diversification	.01	.10	.00	.05	-.02	.24	-.48	.32	
15 Target patents	-.06	-.10	.07	-.05	-.17	.06	-.17	-.04	.20

company’s prior knowledge is utilized in our model. Including the patent scope as a control variable makes our test more conservative. The independent variables that increase explorativeness also correlate negatively with the scope of the patent, thus factors that affect exploration are partially captured by the control. Historical firm-level control for exploitative learning has a significant effect on subsequent learning. This suggests that there is considerable, persistent heterogeneity in organizational routines and strategies related to learning from external sources.

Model 2 presents the effects of external venturing relationship types in comparison to acquisitions. When relational characteristics are not taken into account, both alliances and joint ventures are significantly more likely to lead to explorative outcomes than acquisitions in the model. While CVC has a positive coefficient on exploration, the standard error is relatively large and effects are only weakly significant. Model 3 shows the effects of relational factors on the explorativeness of learning outcomes. Technological relatedness shows a significant effect on increased exploration. Industry relatedness and downstream vertical relatedness fail to show a significant effect.

Model 4 tests our hypotheses with all independent variables entered simultaneously in the regression model. Our first hypothesis predicted that acquisitions would be the

Table 2

Logistic Regression on Explorativeness of Inter-Firm Learning (N = 5,091)

	1	2	3	4	5
Corporate venture capital		1.09 [†] (.71)		.63 (.64)	.62 (.65)
Alliance		1.39*** (.45)		1.03*** (.34)	1.03** (.35)
Joint venture		1.23** (.45)		.85** (.36)	.83** (.36)
Industry relatedness			.26 (.45)	.22 (.37)	
Downstream vertical relatedness			-7.01 [†] (4.70)	-5.14 [†] (3.71)	-3.53* (1.99)
Technology relatedness			-9.58*** (2.48)	-8.10*** (1.99)	-8.33*** (2.11)
Patent scope	-.02*** (.01)	-.02*** (.01)	-.02*** (.01)	-.02*** (.01)	-.02*** (.01)
Previous citations to patent	-.04*** (.01)	-.03** (.01)	-.04*** (.01)	-.03*** (.01)	-.03*** (.01)
Previous explorative inventions (%)	2.09*** (.48)	1.90*** (.41)	1.96*** (.42)	1.89*** (.41)	1.86*** (.41)
Sales (ln)	-.23* (.10)	-.22* (.09)	-.20* (.09)	-.20* (.09)	-.20* (.09)
Research and development spending (%)	-2.81 (3.69)	-1.80 (2.76)	-1.44 (2.96)	-.89 (2.46)	-.86 (2.49)
Technological diversification	-.11 (.53)	.19 (.57)	-.08 (.48)	.09 (.51)	.09 (.51)
Target patents	.00 (.00)	-.01* (.00)	.00 (.00)	.00 (.00)	.00 (.00)
Constant	1.32 (1.38)	.03 (1.40)	1.85 (1.20)	.77 (1.24)	.80 (1.23)

Robust standard errors in parentheses. Year and industry dummies hidden.

[†] $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$. One-sided tests for hypothesized variables, two-sided tests for control variables.

most likely governance mode to lead to exploitative learning outcomes. Joint ventures were predicted to be more explorative than acquisitions, and less explorative than alliances. Corporate venture capital investments were predicted to be the most explorative of governance modes. The results provide partial support for our hypotheses. According to the model, acquisitions are the most likely mode of external corporate venturing to lead to exploitative learning outcomes. Both alliances and joint ventures lead to explorative outcomes by a statistically significant margin compared to acquisitions. However, the effects of venture capital investments remain undetermined due to large variation.

While the regression coefficients suggest that alliances lead to explorative learning outcomes more often than joint ventures, the model does not show whether the difference between alliances and joint ventures is significant. We conducted an additional test to evaluate the difference between joint ventures and alliances (model not shown here). The test did not provide a statistically significant difference. In summary, our findings

support the proposition that less integrated venturing governance modes lead to exploration more than acquisitions, but we are unable to find a statistically significant difference between alliances and joint ventures.

In Hypothesis 2, we predicted that industry relatedness would decrease the likelihood of explorative learning outcomes. Hypothesis 3 suggested that collaboration with companies that operate in industries that are customers to the focal company are more likely to lead to exploitative learning. Models 3 and 4 fail to provide support for these hypotheses. As industry relatedness and vertical downstream vertical relatedness had a high correlation coefficient (.76), we removed each of the variables in turn from the regression. Once industry relatedness was removed, downstream vertical relatedness became significant on the $p < .5$ level (shown in Model 5). Removing downstream vertical relatedness did not provide significant results for industry relatedness. Thus, we found some support for the Hypothesis 3, but had to reject Hypothesis 2.

Finally, Hypothesis 4 predicted that the similarity of partner's technological base would decrease the explorativeness of learning outcomes. Model 4 provides strong support for the hypothesis. In the next chapter, we discuss the findings from theoretical and practical perspectives.

Discussion

In this study, we set out to examine factors influencing explorative versus exploitative learning outcomes from external corporate ventures. In our analysis of interorganizational knowledge flows, as reflected in patent citations between the largest ICT corporations and their external ventures in 1992–2000, we find broad support for our hypotheses concerning the effects of external venture governance modes, the similarity of companies' technological competences, and the downstream vertical relatedness on the type of the learning outcomes.

We find that the less integrated the venture governance mode, the more explorative the learning. In other words, nonequity venturing alliances are likely to be more suitable for explorative learning than joint ventures or acquisitions. Our results related to governance mode can be explained by two mechanisms. The first explanation is related to risk management in corporate decision making. Companies are likely to select a less integrated governance mode such as CVC investment, alliance, or joint venture as the governance mode when they conduct risky explorative ventures. Corporate venture capital and alliances are the least expensive forms of external corporate venturing, and thus they might allow the company to limit its risks. The second explanation is related to differences in the organizational integration of ventures employing different governance modes. In alliances and joint ventures, companies are likely to be less constraint by existing corporate agendas compared to acquisitions that integrates the venture closely into the corporation. By separating ventures from the corporate agenda, explorative learning is fostered while tight integration might foster exploitative learning.

Our empirical tests provided only weakly significant results for the relationship between CVC investment and explorative and exploitative learning outcomes. We inspected our data of CVC investments more closely and found that only 7 CVC investment targets out of 998 led to patents, thus providing observations for our study. Only 105 investment targets had generated any patents. This finding lends itself to two interpretations. One might conclude that technological learning is not taking place in these relationships, perhaps because research and development personnel do not have direct connections with the start-up company and CVC unit is inefficient in absorbing techno-

logical knowledge.³ Alternatively, one might argue that learning through CVC investments is probably not focused on technological learning that would lead to patentable inventions but may instead focus on monitoring the development of technologies, markets, and business models (Block & MacMillan, 1993; Maula et al., 2003).

Concerning relatedness, we found that technological relatedness and downstream vertical relatedness decrease the likelihood of explorative learning. This suggests that companies seeking to enhance their explorative learning need to seek partners with dissimilar technologies and beyond their customers. Surprisingly, our results did not find support for a relationship between industry relatedness and explorative learning. It appears as if technological knowledge bases and industry classifications would be loosely related, and companies would be able to create explorative learning outcomes in ventures within industries, as well as exploitative learning outcomes in ventures across industries. These findings have implications for the growing literature of absorptive capacity and interorganizational learning. Although similarity may improve the quantity of learning (Cohen & Levinthal, 1989; Lane & Lubatkin, 1998), it also leads to enforcement of existing organizational routines, and it may lead to stagnation.

Overall, we make three main contributions in our study. First, we contribute to corporate entrepreneurship literature by developing an interorganizational learning perspective to the study of external corporate ventures. By focusing on a specific set of clearly identifiable activities and their impact on an organizational outcome variable such as organizational learning, our study differs from other perspectives on corporate entrepreneurship that try to identify situations in which an “entrepreneurial philosophy” permeates an organization (Covin & Miles, 1999) or factors that correlate with more entrepreneurial activity of firms (e.g., Miller, 1983; Zahra, 1991). Our study also differs from studies that focus on the level of corporate ventures or initiatives for instance by how individuals champion new ventures (e.g., Dougherty & Hardy, 1996; Dougherty & Heller, 1994) or what factors correlate with success of ventures (e.g., Zahra, 1996). These views of corporate entrepreneurship are not inherently alternatives but complement each other.

Second, our study contributes to interorganizational learning literature by empirically examining factors influencing the type of learning outcomes, namely explorative versus exploitative interorganizational learning, thereby extending the prior literature that has mainly focused on the quantity of learning. Research on organizational learning has repeatedly argued that firms need to strike a balance between exploration and exploitation to ensure long-term survival and maximize performance in the long run (March, 1991). To balance exploration and exploitation, some authors have suggested maintaining a set of organizational activities, each of which contributes to a particular type of corporate renewal in the exploration–exploitation continuum (Burgelman & Sayles, 1986; Keil, 2002). However, previous research has failed to systematically analyze how different external venturing governance modes influence the balance between explorative and exploitative learning. Our results suggest that the different governance modes in external corporate venturing influence explorative or exploitative learning in corporate venturing initiatives. Therefore, the choice of governance mode might be an important parameter for achieving a balance between exploration and exploitation.

Third, we extend the prior literature on interorganizational learning by empirically analyzing the effects of several relational factors on interorganizational learning, thus contributing to the understanding of interorganizational learning. Our results suggest that

3. We are grateful to an anonymous reviewer for pointing this out.

different facets of relatedness between partners in a venture influence not only the quantity of learning as research on absorptive capacity (Cohen & Levinthal, 1990; Lane & Lubatkin, 1998; Zahra & George, 2002) suggests, but they also affect the type of learning taking place.

While we have gone to great lengths and carried out several types of robustness tests to ensure the robustness of our results, some limitations remain that need to be addressed in future research. First, our analysis focused on four key sectors of ICT industries. Given that these sectors are some of the most active sectors in terms of corporate venturing activities, their learning from their external ventures could be different compared to some other industries. Second, in our operationalization of interorganizational learning, we focus only on novel technological innovations. Future research could measure other types of learning outcomes. Third, we concentrated explicitly on the benefits accruing to large companies that collaborate with small- and medium-sized companies.

This study has direct managerial implications. Previous research has implied that companies willing to maximize learning should prefer joint ventures (Mowery, Oxley, & Silverman, 1996) with culturally and technologically similar companies (Lane & Lubatkin, 1998; Simonin, 1999). Alternatively, companies should acquire technologically similar companies to maximize learning (Ahuja & Katila, 2001). Our results imply that managers who follow these guidelines maximize exploitative learning. According to this study, managers willing to maximize explorative learning should prefer alliances and joint ventures with unrelated companies. Our study provides an insight into differences that exist between corporate entrepreneurship initiatives. Some initiatives are more likely to bring incremental benefits that leverage existing competencies; some are more likely to create novel business opportunities that may serve as avenues for future growth. Given that managers are typically advised to balance the mix of explorative and exploitative learning (Eisenhardt & Brown, 1997; March, 1991), the choice between explorative and exploitative focus should depend on the overall strategy of the corporation, as well as the speed of technological change.

To conclude, this article is the first one to empirically link the type of interorganizational learning outcomes (explorative versus exploitative) to the governance modes and relatedness of external ventures. In so doing, we hope that our article stimulates further, more fine-grained research on the various types of learning outcomes in external ventures and interorganizational relationships more generally.

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